

Chapter 3: Denoising Diffusion Probabilistic Models

2026 年 7 月 4 日

Deep Generative Models Notes

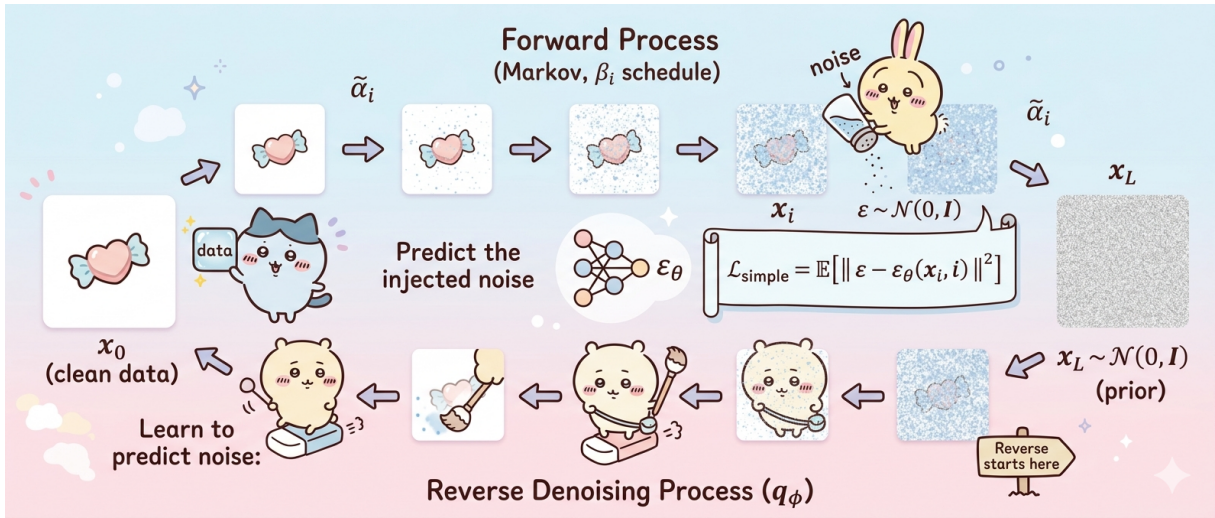
Overview

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1 Overview

Definition 1: DDPM as Two Stochastic Processes

A denoising diffusion probabilistic model (DDPM) is defined by two coupled stochastic processes:

- a **forward process** that gradually corrupts data by Gaussian noise;
- a **reverse denoising process** that learns to remove noise step by step.

The forward chain starts at real data $x_0 \sim p_{\text{data}}$ and moves toward an isotropic Gaussian prior:

$$x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_L, \quad x_L \sim \bar{p}_L, \quad \bar{p}_L \approx p_{\text{prior}} = \mathcal{N}(0, \mathbf{I}).$$

The reverse chain learns a parameterized transition

$$q_\phi(x_{i-1} | x_i), \quad i = L, \dots, 1,$$

to transform unstructured Gaussian noise back to structured samples.

2 Forward Process (Fixed Encoder)

2.1 Markov Corruption Kernel

Definition 2: Forward Transition

Let $\{\beta_i\}_{i=1}^L$ be a predefined monotone increasing schedule. Define

$$\alpha_i \triangleq \sqrt{1 - \beta_i}, \quad i = 1, \dots, L.$$

The forward process is the Markov chain

$$p(x_i | x_{i-1}) = \mathcal{N}(x_i; \alpha_i x_{i-1}, \beta_i \mathbf{I}).$$

Remark 1: Noise Schedule Intuition

Each forward step first scales the previous state by α_i and then injects isotropic Gaussian noise with variance β_i . As i grows, signal shrinks and noise accumulates.

2.2 Closed-Form Marginal at Arbitrary Time

Define the cumulative product

$$\bar{\alpha}_i \triangleq \prod_{k=1}^i \alpha_k = \prod_{k=1}^i \sqrt{1 - \beta_k}.$$

Proposition 1: Closed Form of $p_i(x_i | x_0)$

For every $i \in \{1, \dots, L\}$, the forward law at noise level i given clean x_0 is

$$p_i(x_i | x_0) = \mathcal{N}(x_i; \bar{\alpha}_i x_0, (1 - \bar{\alpha}_i^2) \mathbf{I}).$$

Equivalently,

$$x_i = \bar{\alpha}_i x_0 + \sqrt{1 - \bar{\alpha}_i^2} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I}).$$

Proof. Use induction on i . The statement is immediate for $i = 1$. Assume it holds for $i - 1$. Since

$$x_i = \alpha_i x_{i-1} + \sqrt{\beta_i} \xi_i, \quad \xi_i \sim \mathcal{N}(0, \mathbf{I}),$$

substitute the induction form of x_{i-1} :

$$x_i = \alpha_i \left(\bar{\alpha}_{i-1} x_0 + \sqrt{1 - \bar{\alpha}_{i-1}^2} \varepsilon' \right) + \sqrt{\beta_i} \xi_i = \bar{\alpha}_i x_0 + \eta_i,$$

where η_i is Gaussian with covariance

$$\alpha_i^2 (1 - \bar{\alpha}_{i-1}^2) \mathbf{I} + \beta_i \mathbf{I} = (1 - \bar{\alpha}_i^2) \mathbf{I}.$$

Thus $x_i | x_0$ is Gaussian with the claimed mean and covariance. $\square$

The forward marginal $p_i(x_i | x_0) = \mathcal{N}(\bar{\alpha}_i x_0, (1 - \bar{\alpha}_i^2) \mathbf{I})$ converges to $\mathcal{N}(0, \mathbf{I})$ and becomes independent of x_0 if and only if $\bar{\alpha}_i = \prod_{k=1}^i \sqrt{1 - \beta_k} \rightarrow 0$. A monotone-increasing schedule $\{\beta_i\}$ bounded away from zero is a sufficient (but not necessary) condition for $\bar{\alpha}_L \rightarrow 0$. Integrating

over the unknown data law p_{data} gives the population marginal

$$\bar{p}_i(x_i) = \int p_{\text{data}}(x_0) p_i(x_i | x_0) dx_0,$$

Remark 2: Marginal Levels and Kernel Naming

The population marginal differs at each noise level ($\bar{p}_i \neq \bar{p}_{i'}$ for $i \neq i'$), even though every one-step kernel $p(x_i | x_{i-1})$ is known in closed form. We write p_{data} for the clean-data distribution; $p_i(x_i | x_0)$ for the forward law of x_i given x_0 ; and $\bar{p}_i(x_i) = \int p_{\text{data}}(x_0) p_i(x_i | x_0) dx_0$ for its population marginal. Only adjacent-time maps $p(x_i | x_{i-1})$ are called *transition kernels*. Reverse *kernels* are denoted by q (learned transitions use q_ϕ). This is the opposite of the original DDPM paper (*Ho et al. 2020*), which uses q for the forward process and p_θ for the reverse model.

3 Reverse Denoising Process

3.1 Learning an Approximate Reverse Kernel

The true reverse transition $q(x_{i-1} | x_i)$ is unknown. DDPM learns a tractable Gaussian family:

$$q_\phi(x_{i-1} | x_i) \triangleq \mathcal{N}(x_{i-1}; \mu_\phi(x_i, i), \sigma^2(i)\mathbf{I}),$$

where $\mu_\phi(\cdot, i) : \mathbb{R}^D \rightarrow \mathbb{R}^D$ is learnable and $\sigma^2(i) > 0$ is a fixed schedule. Following the original DDPM paper, $\sigma^2(i)$ is set equal to the forward-posterior variance $\sigma^2(i) = \frac{(1-\bar{\alpha}_{i-1}^2)\beta_i}{1-\bar{\alpha}_i^2}$ (derived in Proposition 2), which ensures the mean-matching KL identity used throughout.

The stage-wise objective is to minimize

$$\mathbb{E}_{x_i \sim \bar{p}_i} [D_{\text{KL}}(q(x_{i-1} | x_i) \| q_\phi(x_{i-1} | x_i))].$$

3.2 Conditioning on x_0

By Bayes' rule under the forward chain,

$$q(x_{i-1} | x_i) = \frac{p(x_i | x_{i-1}) \bar{p}_{i-1}(x_{i-1})}{\bar{p}_i(x_i)}.$$

Each forward kernel $p(x_i | x_{i-1})$ is explicit, but the reverse kernel still depends on the marginals $\bar{p}_{i-1}$ (or p_{data} when $i = 1$) and $\bar{p}_i$. The latter is obtained from p_{data} via

$$\bar{p}_i(x_i) = \int p_{\text{data}}(x_0) p_i(x_i | x_0) dx_0,$$

and p_{data} is unknown in practice. Therefore, even with a fully specified forward process, we cannot evaluate $\bar{p}_i(x_i)$ or $q(x_{i-1} | x_i)$ pointwise. The remedy is to condition on a clean sample $x_0 \sim p_{\text{data}}$, for which all forward quantities are Gaussian:

$$p(x_{i-1} | x_i, x_0) = p(x_i | x_{i-1}) \frac{p_{i-1}(x_{i-1} | x_0)}{p_i(x_i | x_0)}.$$

All three factors on the right are Gaussian, so the posterior above is also Gaussian. Training then replaces expectations under $\bar{p}_i$ by Monte Carlo over $x_0 \sim p_{\text{data}}$ and $x_i \sim p_i(\cdot | x_0)$.

Theorem 1: KL Decomposition for Reverse Matching

For each i , define the marginal and posterior reverse-KL objectives

$$\begin{aligned}\mathcal{J}_i^{\text{rev}}(\phi) &\triangleq \mathbb{E}_{x_i \sim \bar{p}_i} [D_{\text{KL}}(q(x_{i-1} | x_i) \| q_\phi(x_{i-1} | x_i))], \\ \mathcal{J}_i^{\text{post}}(\phi) &\triangleq \mathbb{E}_{x_0 \sim p_{\text{data}}} \mathbb{E}_{x_i \sim p_i(\cdot | x_0)} [D_{\text{KL}}(p(x_{i-1} | x_i, x_0) \| q_\phi(x_{i-1} | x_i))].\end{aligned}$$

Then

$$\mathcal{J}_i^{\text{rev}}(\phi) = \mathcal{J}_i^{\text{post}}(\phi) + C_i,$$

where C_i is independent of ϕ . In particular, minimizing the conditional objective is equivalent to minimizing the marginal reverse-KL up to an additive constant.

Proof. Fix x_i . The true reverse kernel is a mixture of tractable posteriors over x_0 :

$$q(x_{i-1} | x_i) = \int p(x_{i-1} | x_i, x_0) p_{\text{data}}(x_0 | x_i) dx_0 = \mathbb{E}_{x_0 \sim p_{\text{data}}(\cdot | x_i)} [p(x_{i-1} | x_i, x_0)],$$

where $p_{\text{data}}(x_0 | x_i) \propto p_{\text{data}}(x_0) p_i(x_i | x_0)$. Subtract the conditional objective from the marginal KL; the q_ϕ terms cancel:

$$\begin{aligned}&\mathbb{E}_{x_i \sim \bar{p}_i} [D_{\text{KL}}(q(x_{i-1} | x_i) \| q_\phi(x_{i-1} | x_i))] - \mathbb{E}_{x_0 \sim p_{\text{data}}} \mathbb{E}_{x_i \sim p_i(\cdot | x_0)} [D_{\text{KL}}(p(x_{i-1} | x_i, x_0) \| q_\phi(x_{i-1} | x_i))] \\ &= \mathbb{E}_{(x_{i-1}, x_i) \sim p(x_{i-1}, x_i)} [\log q(x_{i-1} | x_i)] - \mathbb{E}_{(x_0, x_{i-1}, x_i) \sim p(x_0, x_{i-1}, x_i)} [\log p(x_{i-1} | x_i, x_0)] \\ &= \mathbb{E}_{x_i \sim \bar{p}_i} \mathbb{E}_{x_0 \sim p_{\text{data}}(\cdot | x_i)} \left[\log p_{\text{data}}(x_0 | x_i) - \mathbb{E}_{x_{i-1} \sim p(\cdot | x_i, x_0)} [\log p_{\text{data}}(x_0 | x_{i-1}, x_i)] \right],\end{aligned}$$

using $\log q(x_{i-1} | x_i) - \log p(x_{i-1} | x_i, x_0) = \log p_{\text{data}}(x_0 | x_i) - \log p_{\text{data}}(x_0 | x_{i-1}, x_i)$, where $p(x_{i-1}, x_i) = p(x_i | x_{i-1}) \bar{p}_{i-1}(x_{i-1})$ and $p(x_0, x_{i-1}, x_i) = p_{\text{data}}(x_0) p(x_{i-1} | x_i, x_0) p_i(x_i | x_0)$. This depends only on the fixed forward chain, hence it is a ϕ -free constant C_i . Therefore $\mathcal{J}_i^{\text{rev}}(\phi) = \mathcal{J}_i^{\text{post}}(\phi) + C_i$. $\square$

Remark 3: Why Conditional Training Is Legitimate (Two Independent Routes)

The stage-wise goal is to match the *marginal* reverse kernel $q(x_{i-1} | x_i)$, but this object depends on unknown marginals $\bar{p}_{i-1}, \bar{p}_i$ and cannot be evaluated directly. Conditioning on a clean sample x_0 replaces it with the tractable posterior $p(x_{i-1} | x_i, x_0)$.

Route A (Theorem 1 only). Theorem 1 shows $\mathcal{J}_i^{\text{rev}}(\phi) = \mathcal{J}_i^{\text{post}}(\phi) + C_i$ with C_i independent of ϕ . So minimizing the conditional objective is exactly the same as minimizing the marginal reverse-KL up to an additive constant. This route does *not* require the ELBO of Section 4.

Route B (original DDPM paper). The paper instead starts from maximizing $\log q_\phi(x_0)$ and derives a variational upper bound (Theorem 2), whose diffusion term already has the conditional form $D_{\text{KL}}(p(x_{i-1} | x_i, x_0) \| q_\phi(x_{i-1} | x_i))$. That proof is self-contained and does not use Theorem 1.

Both routes justify the same practical recipe—sample $x_0 \sim p_{\text{data}}$, corrupt to x_i , and train q_ϕ against the Gaussian forward posterior—but they answer the question at different levels: Route A is step-wise reverse matching; Route B is end-to-end likelihood maximization.

Proposition 2: Closed Form of $p(x_{i-1} \mid x_i, x_0)$

The conditional posterior is

$$p(x_{i-1} \mid x_i, x_0) = \mathcal{N}(x_{i-1}; \mu(x_i, x_0, i), \sigma^2(i)\mathbf{I}),$$

with

$$\begin{aligned} \mu(x_i, x_0, i) &= \frac{\alpha_i - \bar{\alpha}_i^2/\alpha_i}{1 - \bar{\alpha}_i^2} x_i + \frac{(1 - \alpha_i^2)\bar{\alpha}_i/\alpha_i}{1 - \bar{\alpha}_i^2} x_0, \\ \sigma^2(i) &= \frac{1 - \bar{\alpha}_{i-1}^2}{1 - \bar{\alpha}_i^2} \beta_i. \end{aligned}$$

Proof. This is the adjacent-step special case of the general reverse conditional transition formula for linear Gaussian corruptions; the general case for arbitrary $0 \leq t < s$ will be discussed later. We prove the one-step DDPM case directly.

Conditioned on x_0 , the forward marginal and one-step transition are

$$p_{i-1}(x_{i-1} \mid x_0) = \mathcal{N}(x_{i-1}; \bar{\alpha}_{i-1}x_0, (1 - \bar{\alpha}_{i-1}^2)\mathbf{I}), \quad p(x_i \mid x_{i-1}) = \mathcal{N}(x_i; \alpha_i x_{i-1}, \beta_i \mathbf{I}).$$

By Bayes' rule,

$$p(x_{i-1} \mid x_i, x_0) \propto p(x_i \mid x_{i-1}) p_{i-1}(x_{i-1} \mid x_0).$$

As a function of x_{i-1} , the log-density is quadratic, so the posterior is Gaussian. Its precision is

$$\frac{\alpha_i^2}{\beta_i} + \frac{1}{1 - \bar{\alpha}_{i-1}^2} = \frac{1 - \bar{\alpha}_i^2}{\beta_i(1 - \bar{\alpha}_{i-1}^2)},$$

hence

$$\sigma^2(i) = \frac{\beta_i(1 - \bar{\alpha}_{i-1}^2)}{1 - \bar{\alpha}_i^2}.$$

The corresponding mean is

$$\mu(x_i, x_0, i) = \sigma^2(i) \left(\frac{\alpha_i}{\beta_i} x_i + \frac{\bar{\alpha}_{i-1}}{1 - \bar{\alpha}_{i-1}^2} x_0 \right) = \frac{\alpha_i(1 - \bar{\alpha}_{i-1}^2)}{1 - \bar{\alpha}_i^2} x_i + \frac{\beta_i \bar{\alpha}_{i-1}}{1 - \bar{\alpha}_i^2} x_0.$$

Since $\bar{\alpha}_{i-1} = \bar{\alpha}_i/\alpha_i$ and $\beta_i = 1 - \alpha_i^2$, this is equivalently

$$\mu(x_i, x_0, i) = \frac{\alpha_i - \bar{\alpha}_i^2/\alpha_i}{1 - \bar{\alpha}_i^2} x_i + \frac{(1 - \alpha_i^2)\bar{\alpha}_i/\alpha_i}{1 - \bar{\alpha}_i^2} x_0,$$

which proves the claimed form. $\square$

This yields the diffusion training term (mean-matching form, already averaged over x_0):

$$\mathcal{L}_{\text{diffusion}}(\phi) \triangleq \sum_{i=2}^L \frac{1}{2\sigma^2(i)} \mathbb{E}_{x_0 \sim p_{\text{data}}} \mathbb{E}_{x_i \sim p_i(\cdot \mid x_0)} \left[\|\mu_\phi(x_i, i) - \mu(x_i, x_0, i)\|_2^2 \right].$$

Because q_ϕ and $p(x_{i-1} \mid x_i, x_0)$ share the same variance $\sigma^2(i)$, the Gaussian KL satisfies

$$D_{\text{KL}}(p(x_{i-1} \mid x_i, x_0) \parallel q_\phi(x_{i-1} \mid x_i)) = \frac{1}{2\sigma^2(i)} \|\mu(x_i, x_0, i) - \mu_\phi(x_i, i)\|_2^2.$$

Therefore,

$$\mathcal{L}_{\text{diffusion}}(\phi) = \mathbb{E}_{x_0 \sim p_{\text{data}}} [\mathcal{L}_{\text{diffusion}}(x_0; \phi)],$$

where $\mathcal{L}_{\text{diffusion}}(x_0; \phi)$ is the per-sample KL sum appearing in the ELBO (Section 4).

4 DDPM ELBO

4.1 Joint Model and Variational Bound

Define the latent trajectory $x_{1:L}$ and generative joint

$$q_\phi(x_0, x_{1:L}) = p_{\text{prior}}(x_L) \prod_{i=1}^L q_\phi(x_{i-1} | x_i),$$

where $p_{\text{prior}}(x_L) = \mathcal{N}(0, \mathbf{I})$ is the sampling prior at the top of the reverse chain (the forward marginal satisfies $\bar{p}_L \approx p_{\text{prior}}$ by schedule design). Let the forward trajectory given x_0 be Markov with transitions $p(x_i | x_{i-1})$, so

$$\prod_{i=1}^L p(x_i | x_{i-1})$$

is the joint density of $(x_1, \dots, x_L)$ and $\bar{p}_i(x_i) = \int p_{\text{data}}(x_0) p_i(x_i | x_0) dx_0$.

Theorem 2: ELBO Decomposition for DDPM

For any data sample x_0 ,

$$-\log q_\phi(x_0) \leq -\mathcal{L}_{\text{ELBO}}(x_0; \phi) = \mathcal{L}_{\text{prior}}(x_0) + \mathcal{L}_{\text{recon}}(x_0; \phi) + \mathcal{L}_{\text{diffusion}}(x_0; \phi),$$

with

$$\begin{aligned} \mathcal{L}_{\text{prior}}(x_0) &\triangleq D_{\text{KL}}(p_L(\cdot | x_0) \| p_{\text{prior}}(x_L)), \\ \mathcal{L}_{\text{recon}}(x_0; \phi) &\triangleq \mathbb{E}_{x_1 \sim p_1(\cdot | x_0)} [-\log q_\phi(x_0 | x_1)], \\ \mathcal{L}_{\text{diffusion}}(x_0; \phi) &\triangleq \sum_{i=2}^L \mathbb{E}_{x_i \sim p_i(\cdot | x_0)} [D_{\text{KL}}(p(x_{i-1} | x_i, x_0) \| q_\phi(x_{i-1} | x_i))]. \end{aligned}$$

Proof. Fix x_0 . The forward joint factorises as

$$p(x_{1:L} | x_0) = \prod_{i=1}^L p(x_i | x_{i-1}).$$

Write $\mathbb{E}_{x_0}^{\text{fwd}}[\cdot] \triangleq \mathbb{E}_{x_{1:L} \sim p(x_{1:L} | x_0)}[\cdot]$ for the expectation over the full forward trajectory given this fixed x_0 . Then

$$\begin{aligned} -\log q_\phi(x_0) &= -\log \int q_\phi(x_0, x_{1:L}) dx_{1:L} \\ &= -\log \int p(x_{1:L} | x_0) \frac{q_\phi(x_0, x_{1:L})}{p(x_{1:L} | x_0)} dx_{1:L} \\ &= -\log \mathbb{E}_{x_0}^{\text{fwd}} \left[\frac{q_\phi(x_0, x_{1:L})}{p(x_{1:L} | x_0)} \right] \\ &\leq \mathbb{E}_{x_0}^{\text{fwd}} \left[-\log q_\phi(x_0, x_{1:L}) + \sum_{i=1}^L \log p(x_i | x_{i-1}) \right] \triangleq \mathcal{B}(x_0; \phi), \end{aligned}$$

by Jensen's inequality ($-\log$ is convex). Using $q_\phi(x_0, x_{1:L}) = p_{\text{prior}}(x_L) \prod_{i=1}^L q_\phi(x_{i-1} | x_i)$,

expand the bound:

$$\begin{aligned}
\mathcal{B}(x_0; \phi) &= \mathbb{E}_{x_0}^{\text{fwd}} \left[-\log p_{\text{prior}}(x_L) + \sum_{i=1}^L \log p(x_i | x_{i-1}) \right] - \mathbb{E}_{x_0}^{\text{fwd}} \left[\sum_{i=1}^L \log q_{\phi}(x_{i-1} | x_i) \right] \\
&= \mathbb{E}_{x_L \sim p_L(\cdot | x_0)} [-\log p_{\text{prior}}(x_L)] \\
&\quad + \mathbb{E}_{x_1 \sim p_1(\cdot | x_0)} [\log p_1(x_1 | x_0) - \log q_{\phi}(x_0 | x_1)] \\
&\quad + \sum_{i=2}^L \mathbb{E}_{(x_{i-1}, x_i) \sim p(\cdot | x_0)} [\log p(x_i | x_{i-1}) - \log q_{\phi}(x_{i-1} | x_i)].
\end{aligned}$$

The middle bracket equals $\mathbb{E}_{x_1 \sim p_1(\cdot | x_0)} [\log p_1(x_1 | x_0)] + \mathcal{L}_{\text{recon}}(x_0; \phi)$; the first part will be cancelled by the telescoping sum below. For each $i \geq 2$, apply Bayes' rule on the forward chain conditioned on x_0 :

$$\log p(x_i | x_{i-1}) = \log p(x_{i-1} | x_i, x_0) + \log p_i(x_i | x_0) - \log p_{i-1}(x_{i-1} | x_0).$$

Hence

$$\begin{aligned}
&\mathbb{E}_{(x_{i-1}, x_i) \sim p(\cdot | x_0)} [\log p(x_i | x_{i-1}) - \log q_{\phi}(x_{i-1} | x_i)] \\
&= \mathbb{E}_{x_i \sim p_i(\cdot | x_0)} [D_{\text{KL}}(p(x_{i-1} | x_i, x_0) \| q_{\phi}(x_{i-1} | x_i))] \\
&\quad + \mathbb{E}_{x_i \sim p_i(\cdot | x_0)} [\log p_i(x_i | x_0)] - \mathbb{E}_{x_{i-1} \sim p_{i-1}(\cdot | x_0)} [\log p_{i-1}(x_{i-1} | x_0)].
\end{aligned}$$

Summing the last two terms from $i = 2$ to L telescopes:

$$\begin{aligned}
&\sum_{i=2}^L \left(\mathbb{E}_{x_i \sim p_i(\cdot | x_0)} [\log p_i(x_i | x_0)] - \mathbb{E}_{x_{i-1} \sim p_{i-1}(\cdot | x_0)} [\log p_{i-1}(x_{i-1} | x_0)] \right) \\
&= \mathbb{E}_{x_L \sim p_L(\cdot | x_0)} [\log p_L(x_L | x_0)] - \mathbb{E}_{x_1 \sim p_1(\cdot | x_0)} [\log p_1(x_1 | x_0)].
\end{aligned}$$

Adding the $i = 1$ bracket (which contained the extra $\mathbb{E}_{x_1 \sim p_1(\cdot | x_0)} [\log p_1(x_1 | x_0)]$ term) to the lower end of the telescope cancels that term, leaving

$$\begin{aligned}
\mathcal{B}(x_0; \phi) &= \mathbb{E}_{x_L \sim p_L(\cdot | x_0)} [\log p_L(x_L | x_0) - \log p_{\text{prior}}(x_L)] \\
&\quad + \mathcal{L}_{\text{recon}}(x_0; \phi) + \sum_{i=2}^L \mathbb{E}_{x_i \sim p_i(\cdot | x_0)} [D_{\text{KL}}(p(x_{i-1} | x_i, x_0) \| q_{\phi}(x_{i-1} | x_i))].
\end{aligned}$$

The first bracket is $\mathcal{L}_{\text{prior}}(x_0)$ and the sum is $\mathcal{L}_{\text{diffusion}}(x_0; \phi)$, hence

$$-\log q_{\phi}(x_0) \leq \mathcal{L}_{\text{prior}}(x_0) + \mathcal{L}_{\text{recon}}(x_0; \phi) + \mathcal{L}_{\text{diffusion}}(x_0; \phi).$$

Taking $\mathbb{E}_{x_0 \sim p_{\text{data}}}[\cdot]$ and using the Gaussian KL identity $D_{\text{KL}}(p(x_{i-1} | x_i, x_0) \| q_{\phi}(x_{i-1} | x_i)) = \frac{1}{2\sigma^2(i)} \|\mu(x_i, x_0, i) - \mu_{\phi}(x_i, i)\|_2^2$ recovers the mean-matching objective $\mathcal{L}_{\text{diffusion}}(\phi)$ of Section 3. $\square$

Remark 4: Role of Three Terms

- $\mathcal{L}_{\text{prior}}$: controlled by the noise schedule so that $p_L(\cdot | x_0) \approx p_{\text{prior}}$ and $\bar{p}_L \approx \mathcal{N}(0, \mathbf{I})$.
- $\mathcal{L}_{\text{recon}}$: likelihood at the first reverse step.
- $\mathcal{L}_{\text{diffusion}}$: matches learned reverse transitions $q_\phi(x_{i-1} | x_i)$ to forward posteriors $p(x_{i-1} | x_i, x_0)$.

5 Practical Choices of Prediction and Loss

5.1 ε -Prediction Parameterization

From the forward closed form,

$$x_i = \bar{\alpha}_i x_0 + \sqrt{1 - \bar{\alpha}_i^2} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I}),$$

the posterior mean can be rewritten as

$$\mu(x_i, x_0, i) = \frac{1}{\alpha_i} \left(x_i - \frac{\beta_i}{\sqrt{1 - \bar{\alpha}_i^2}} \varepsilon \right).$$

This motivates parameterizing

$$\mu_\phi(x_i, i) = \frac{1}{\alpha_i} \left(x_i - \frac{\beta_i}{\sqrt{1 - \bar{\alpha}_i^2}} \varepsilon_\phi(x_i, i) \right),$$

where ε_ϕ directly predicts injected noise.

Remark 5: Noise-Detection View

The network is trained as a noise detector: infer the forward noise from a corrupted sample x_i , then use that estimate to form reverse denoising steps.

5.2 Simplified Objective

Under the above parameterization, the mean-matching objective is equivalent (up to constants and positive time-dependent weights) to weighted noise regression:

$$\|\mu_\phi(x_i, i) - \mu(x_i, x_0, i)\|_2^2 \propto \|\varepsilon_\phi(x_i, i) - \varepsilon\|_2^2.$$

The commonly used simplified loss is

$$\mathcal{L}_{\text{simple}}(\phi) \triangleq \mathbb{E}_{i \sim \text{Uniform}\{1, \dots, L\}} \mathbb{E}_{x_0 \sim p_{\text{data}}} \mathbb{E}_{\varepsilon \sim \mathcal{N}(0, \mathbf{I})} \left[\|\varepsilon_\phi(x_i, i) - \varepsilon\|_2^2 \right],$$

where

$$x_i = \bar{\alpha}_i x_0 + \sqrt{1 - \bar{\alpha}_i^2} \varepsilon.$$

Because this is a least-squares problem, the optimal predictor at each (x_i, i) is the conditional mean:

$$\varepsilon_\phi^*(x_i, i) = \mathbb{E}_{x_0 \sim p_{\text{data}}(\cdot | x_i)} \left[\frac{x_i - \bar{\alpha}_i x_0}{\sqrt{1 - \bar{\alpha}_i^2}} \right], \quad x_i \sim \bar{p}_i,$$

equivalently $\mathbb{E}_{x_0 \sim p_{\text{data}}, \varepsilon \sim \mathcal{N}(0, \mathbf{I})} [\varepsilon | x_i]$ under the forward sampling $x_i = \bar{\alpha}_i x_0 + \sqrt{1 - \bar{\alpha}_i^2} \varepsilon$. Even though a single x_i does not uniquely determine x_0 , averaging over $p_{\text{data}}(x_0 | x_i)$ yields the MSE-optimal noise estimate.

5.3 x_0 -Prediction Parameterization

An equivalent choice is to predict a clean sample directly:

$$x_\phi(x_i, i) \approx x_0.$$

Using Proposition 2, replace x_0 in the posterior mean by $x_\phi(x_i, i)$:

$$\mu_\phi(x_i, i) = \frac{\alpha_i - \bar{\alpha}_i^2/\alpha_i}{1 - \bar{\alpha}_i^2} x_i + \frac{(1 - \alpha_i^2)\bar{\alpha}_i/\alpha_i}{1 - \bar{\alpha}_i^2} x_\phi(x_i, i).$$

Therefore

$$\mu_\phi(x_i, i) - \mu(x_i, x_0, i) = \frac{(1 - \alpha_i^2)\bar{\alpha}_i/\alpha_i}{1 - \bar{\alpha}_i^2} (x_\phi(x_i, i) - x_0),$$

and the diffusion KL term is equivalent to weighted clean-sample regression:

$$\mathcal{L}_{\text{diffusion}}(\phi) = \sum_{i=2}^L w_i \mathbb{E}_{x_0 \sim p_{\text{data}}} \mathbb{E}_{x_i \sim p_i(\cdot | x_0)} [\|x_\phi(x_i, i) - x_0\|_2^2],$$

for positive weights

$$w_i = \frac{1}{2\sigma^2(i)} \left(\frac{(1 - \alpha_i^2)\bar{\alpha}_i/\alpha_i}{1 - \bar{\alpha}_i^2} \right)^2.$$

Hence the least-squares optimal predictor is

$$x_\phi^*(x_i, i) = \mathbb{E}[x_0 | x_i].$$

5.4 Equivalence of Prediction Targets

The targets (μ, x_0, ε) are linearly equivalent. From

$$x_i = \bar{\alpha}_i x_0 + \sqrt{1 - \bar{\alpha}_i^2} \varepsilon,$$

we have

$$x_0 = \frac{x_i - \sqrt{1 - \bar{\alpha}_i^2} \varepsilon}{\bar{\alpha}_i}, \quad \varepsilon = \frac{x_i - \bar{\alpha}_i x_0}{\sqrt{1 - \bar{\alpha}_i^2}}.$$

Hence prediction heads are bijectively related:

$$x_\phi(x_i, i) = \frac{x_i - \sqrt{1 - \bar{\alpha}_i^2} \varepsilon_\phi(x_i, i)}{\bar{\alpha}_i}, \quad \varepsilon_\phi(x_i, i) = \frac{x_i - \bar{\alpha}_i x_\phi(x_i, i)}{\sqrt{1 - \bar{\alpha}_i^2}}.$$

So one may train with either x_0 -prediction or ε -prediction.

6 Sampling from a Trained DDPM

After training $q_\phi(x_{i-1} | x_i)$, generate samples by reverse-time recursion. Start from

$$x_L \sim p_{\text{prior}} = \mathcal{N}(0, \mathbf{I}),$$

then for $i = L, L-1, \dots, 1$, sample

$$x_{i-1} = \mu_\phi(x_i, i) + \sigma(i) z_i, \quad z_i \sim \mathcal{N}(0, \mathbf{I}),$$

with $z_1 \equiv 0$ for the last step.

(a) **ε -prediction form.** Using $\mu_\phi(x_i, i) = \frac{1}{\alpha_i} \left(x_i - \frac{\beta_i}{\sqrt{1-\bar{\alpha}_i^2}} \varepsilon_\phi(x_i, i) \right)$,

$$x_{i-1} = \frac{1}{\alpha_i} \left(x_i - \frac{\beta_i}{\sqrt{1-\bar{\alpha}_i^2}} \varepsilon_\phi(x_i, i) \right) + \sigma(i) z_i.$$

(b) **x_0 -prediction form.** Using $x_\phi(x_i, i)$ instead of ε_ϕ ,

$$x_{i-1} = \frac{\alpha_i - \bar{\alpha}_i^2/\alpha_i}{1 - \bar{\alpha}_i^2} x_i + \frac{(1 - \alpha_i^2)\bar{\alpha}_i/\alpha_i}{1 - \bar{\alpha}_i^2} x_\phi(x_i, i) + \sigma(i) z_i.$$

Both forms are equivalent under the linear map in the previous subsection.

Remark 6: Denoise-Then-Re-noise View

Each reverse step can be interpreted as:

- **Denoise:** use the network prediction from x_i (either x_ϕ or ε_ϕ) to estimate the clean structure;
- **Re-noise:** inject calibrated Gaussian uncertainty through $\sigma(i)z_i$ to obtain x_{i-1} .

This is exactly the DDPM posterior-sampling mechanism under the learned mean.